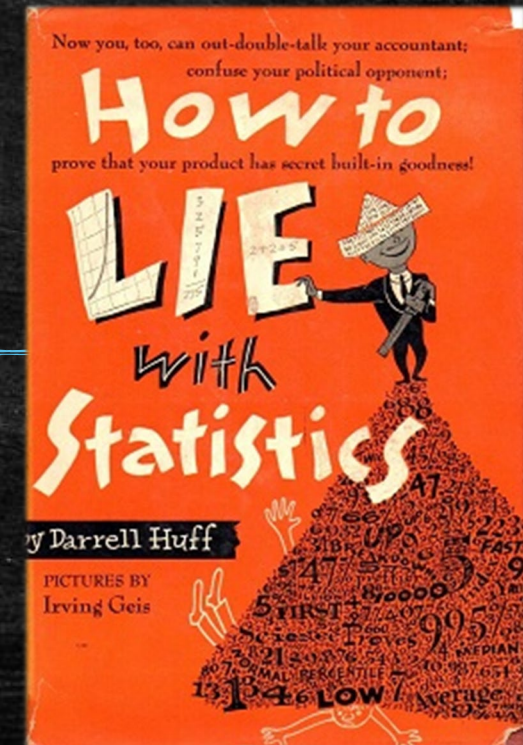
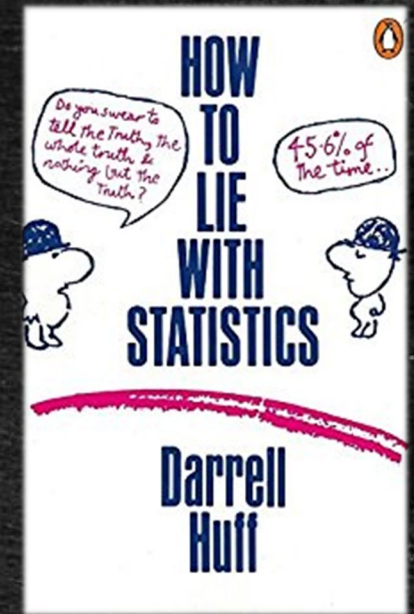
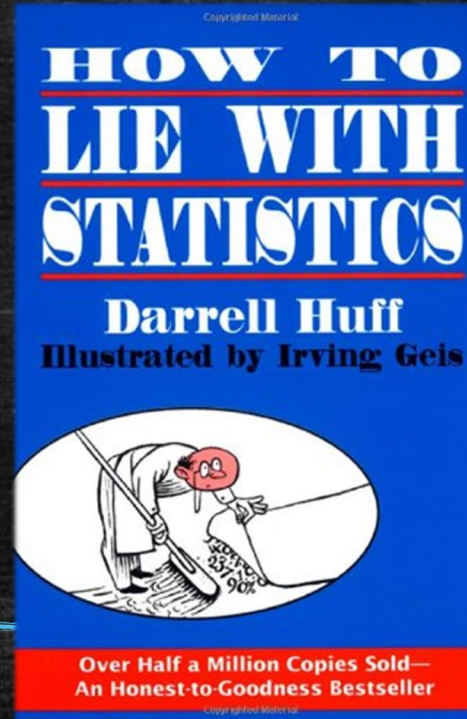
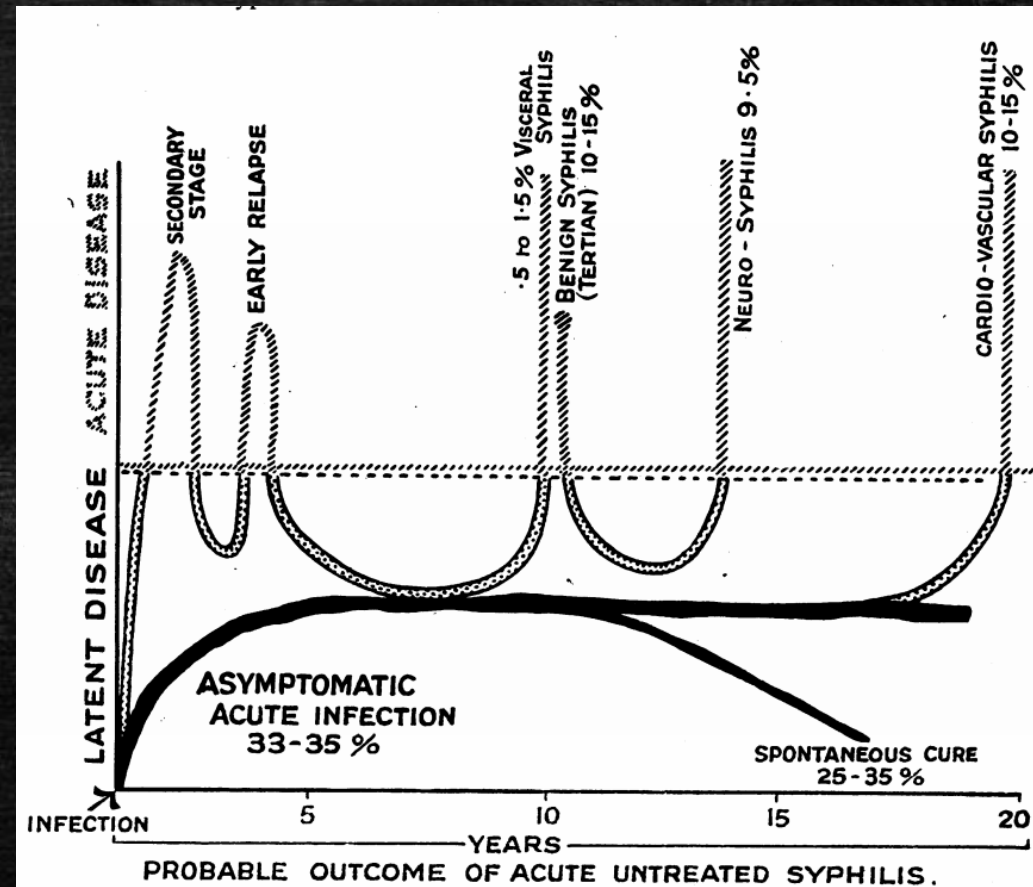


Lying with Statistics



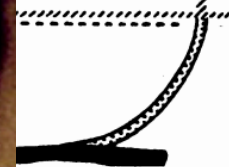
Tuskegee Syphilis Study

- 1926: Syphilis is a major health problem.
 - “one of medicine's most intricate problems”
 - (Jones, 1944)
 - Prevalence of 35 percent in adults





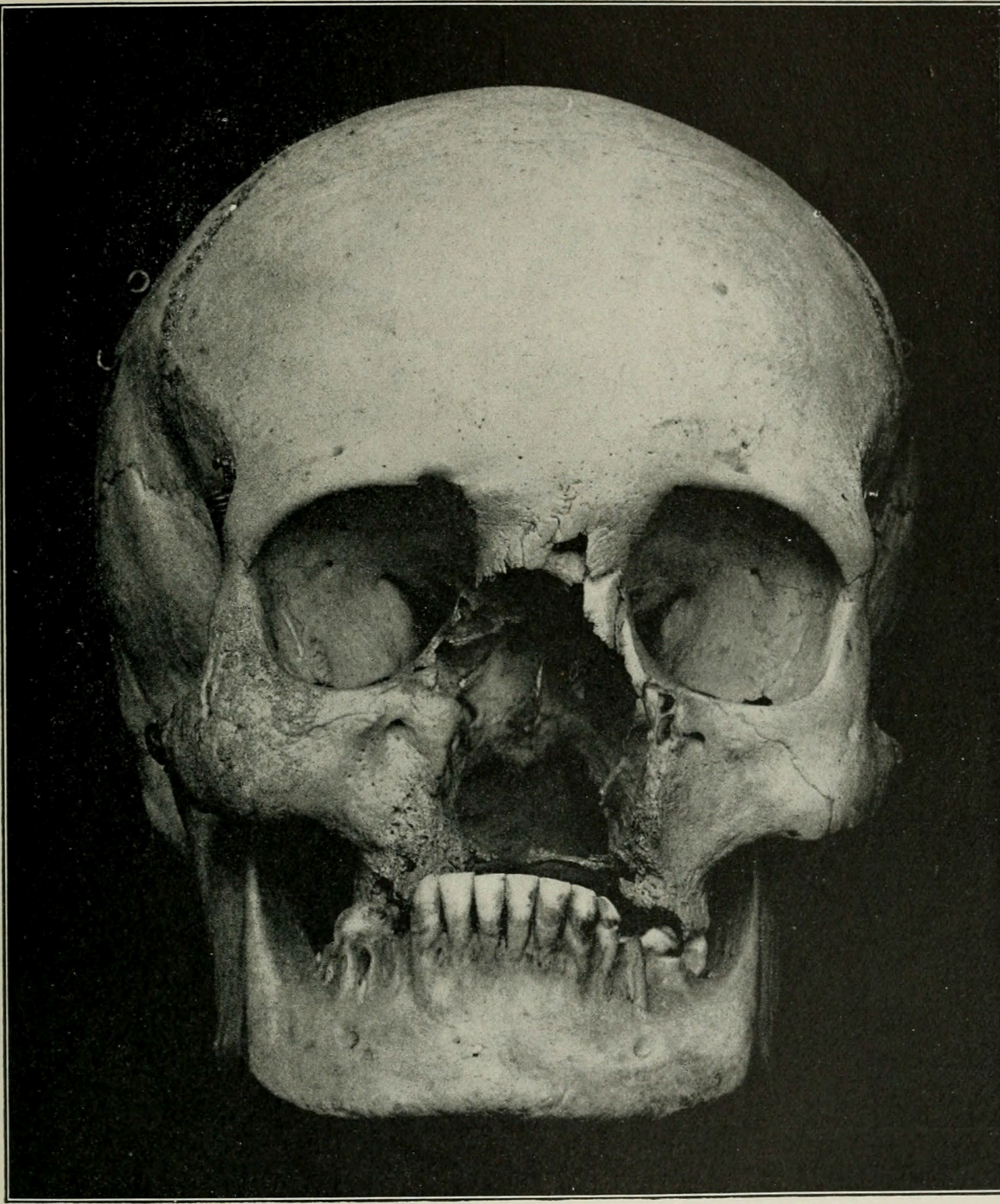
CARDIO-VASCULAR SYPHILIS 10-15 %



SPONTANEOUS CURE
25-35 %

20

SYPHILIS.



CARDIO-VASCULAR SYPHILIS 10-15 %

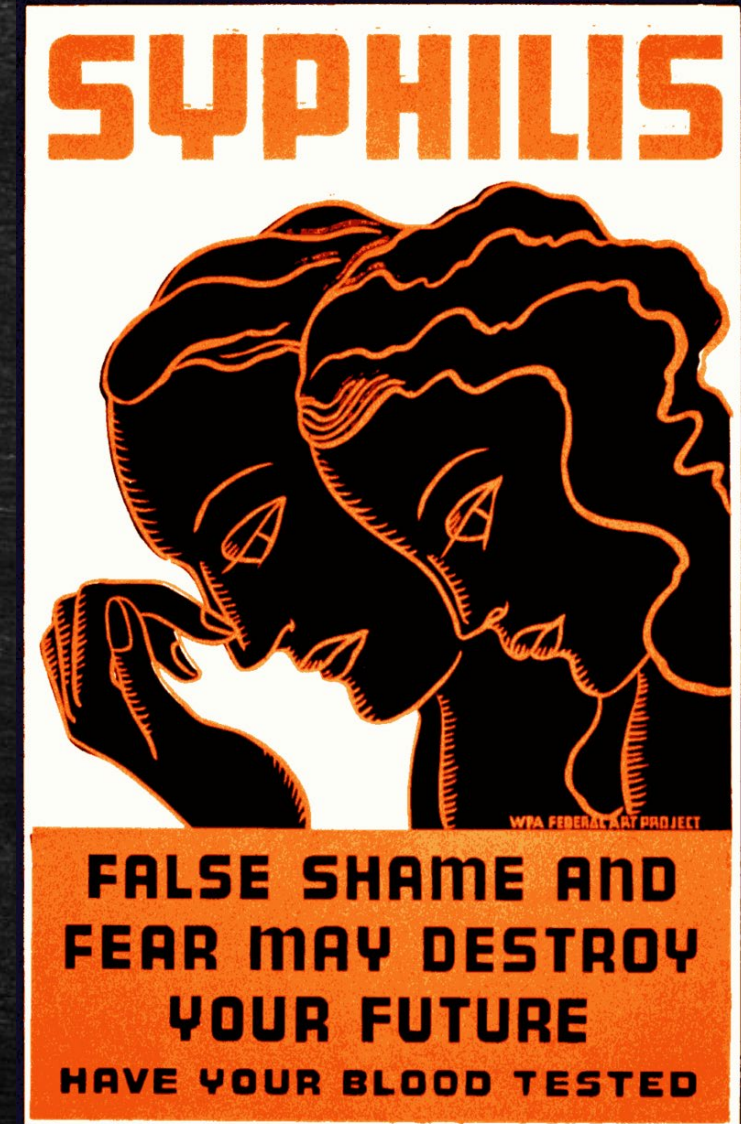
SPONTANEOUS CURE 25-35 %

20

SYPHILIS.

Tuskegee Syphilis Study

- 1929: Aggressive treatment approach
 - initiated with mercury and bismuth.
- Cure rate is less than 30 percent;
 - treatment requires months and
 - side effects are toxic, sometimes fatal.
- 1931: Funding is cut from project
 - decide to follow men left untreated due to lack of funds in order to show need for treatment program.
- 1932: Follow-up effort organized into study of 399 men with syphilis and 201 without.



Tuskegee Syphilis Study

- The men were impoverished, African-American sharecroppers from Alabama
- The men were given periodic physical assessments and
 - told they were being treated for “bad blood,”
 - local term used to describe several ailments, including syphilis, anemia, and fatigue
- Original study was designed to
 - last 6 months

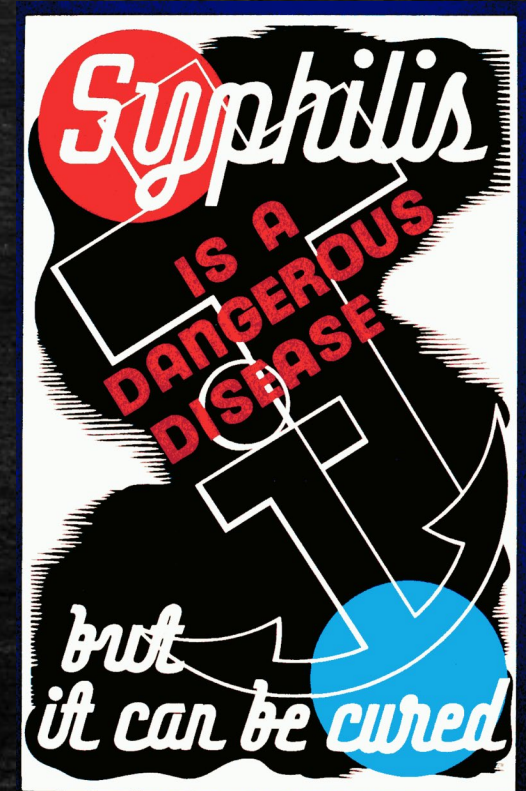
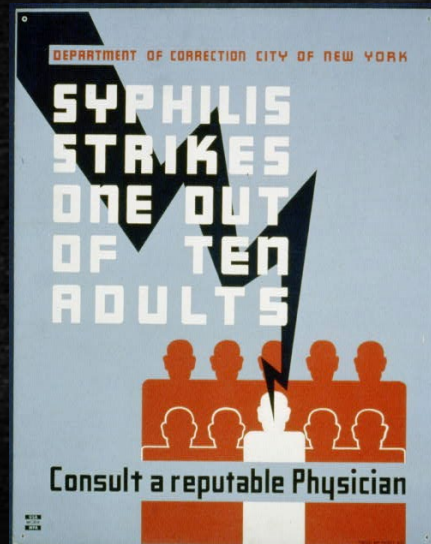
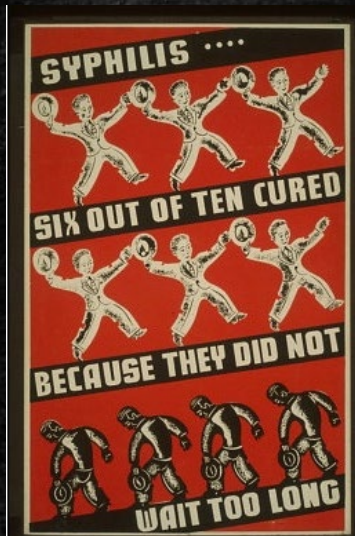


Tuskegee Syphilis Study

- 1936: Major paper published.
 - Study criticized because it is not known if men are being treated.
 - Local physicians asked to assist with study and not to treat men.
 - Decision was made to follow the men until death.
- World War II: Researchers attempted to prevent drafted men in the study from getting treatment
 - "So far, we are keeping the known positive patients from getting treatment."
 - Despite this, 96% of the 90 original test subjects reexamined in 1963
 - had received treatments from another health provider

Tuskegee Syphilis Study

- **1945:** Penicillin accepted as treatment of choice for syphilis.
- **1947:** USPHS establishes “Rapid Treatment Centers” to treat syphilis;
 - men in study are not treated



Tuskegee Syphilis Study

- 1968: 60% of Men in Study are Dead
 - Concerns raised about ethics of study by
 - Peter Buxtun, the eventual "whistleblower", as well as others
- 1969 CDC reaffirms need for study and
 - gains official support from
 - American Medical Association and
 - National Medical Association
- Early 1970s Buxtun leaks story to media
 - The story first reported by
 - Washington Star on July 25, 1972

<u>Classification of Cases in Tuskegee Study</u>			
	Controls	Syphilitic	Total
Classification at initial examination	200	411	611
Cases added in 1938-1939	-	14	14
Total - Original classification	200	425	625
Controls infected during observation	-9	+9	-
Controls reclassified as syphilitic on basis of additional history	-1	+1	-
on basis of treponemal tests	-8	+8	-
Total - Final classification	182	443	625
Known dead - Number	97	276	373
Percent	53.3	62.3	59.7
Remainder -	85	167	252
Examined in 1968			
Number	36	53	89
Percent	42.4	31.7	35.3

Tuskegee Syphilis Study

- July 1972: Ad Hoc Advisory Panel created by US Gov
 - nine members from medicine, law, religion, labor, education, health administration, and public affairs.
- October 1972: The advisory panel concluded that the Tuskegee Study was “ethically unjustified”
- November 1972: Study Officially ends.
 - 74 men in study alive. Of the original 399 infected men,
 - 128 had died of syphilis or related complications,
 - 40 of their wives had been infected, and
 - 19 of their children were born with congenital syphilis.
- **1973** Congress holds hearings and
 - a class-action lawsuit is filed on behalf of the study participants.

Tuskegee Syphilis Study

- 1974 Congress passed the National Research Act
 - and created a commission to study and write regulations governing studies involving human participants.
 - Within the US Department of Health and Human Services,
 - the Office for Human Research Protections (OHRP) was established to oversee clinical trials.
- Now studies require informed consent,
 - communication of diagnosis,
 - accurate reporting of test results, and
 - approval from Institutional review boards (IRBs),
 - including laypeople, are established in scientific research groups and hospitals to review study protocols and protect patient interests, to ensure that participants are fully informed.

Tuskegee Syphilis Study

- 1974: A \$10 million out-of-court settlement is reached and the U.S. government promised to give lifetime medical benefits and burial services to all living participants.
- 1975: Wives, widows, and offspring were added to the program.
- 1997: On May 16th President Clinton apologizes on behalf of the Nation.

Wrapping Up...
